Maheer Ihtejaz Moin

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EDUCATION

The University of Sydney, Sydney, NSW Feb 2024 – Jun 2028 (Expected)
Bachelor of Advanced Computing (Honours) | Majors: Computational Data Science; Finance
Dalyell Scholar (University of Sydney high achievers program); Vice Chancellor's Scholarship

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, SQL, R
Software Engineering: Object-oriented programming, data structures and algorithms, systems architecture, REST API design, concurrency, distributed computing, high-performance and low-latency systems, hashing and probabilistic data structures, code review and refactoring, debugging and profiling, Agile
Cloud, Platforms and Tooling: Linux, Git/GitHub, Docker, Kubernetes, CI/CD (GitHub Actions), Vercel, real-time monitoring and observability dashboards, Pytest, unit, regression and reconciliation testing
Web and Data: React, Node.js, Tailwind CSS, Supabase, PostgreSQL, relational schema design, row-level security, object storage
Quantitative and Machine Learning: NumPy, Pandas, SciPy, time-series forecasting (vector autoregression, GJR-GARCH, HAR-RV), ridge and elastic-net regularisation, Kalman filters, LSTM, XGBoost, Monte Carlo simulation, hypothesis and permutation testing, backtesting

PROFESSIONAL EXPERIENCE

2A Marketing Pty Ltd, Sydney, NSW | **Full-Stack Developer Intern** Mar 2025 – Dec 2025

- Built the company's first online sales platform. Developed a React application with Vite, client-side routing and automatic deployemnts to Vercel through github and routed through the company's domain
- Built a Supabase and PostgreSQL backend that allows non technical staff to manage website listings themselves including authentication, role based access control and row level security
- Developed a document and shipping tracking app giving staff and customers versioned files and real-time updates
- Built an admin dashboard for vehicle listings, including image uploads to object storage, public URL generation and database updates with optimistic UI interactions
- Owned the stack end to end as the sole developer, from database and access-control design through deployment and post-launch maintenance

COMPETITIONS AND AWARDS

IMC Trading AlgoJam, Algorithmic Trading Competition 2026
Team Lead | 9th of 40 teams, top 10 finish | Python, NumPy, Pandas

- Led the team's research and development of the final trading system, tested strategies against unseen market data and achieved $568,609 in backtest PnL with a 14.85 Sharpe ratio and $4,898 maximum drawdown across nine instruments implementing it as a dependency-free system running on stock CPython with no third-party packages.
- Improved performance on the game-theoretic instrument to $588,237 mean PnL across 300 held out simulations by modeling the room's previous day positioning and only trading when the estimated edge was statistically meaningful
- Held zero risk-limit breaches, invalid positions or type errors across 365 days, 1,056 Monte Carlo scenarios and adversarial price paths, and retired three signals fitted to noise and two strategy layers worth $152,800 a year, by replacing a static 3% margin with exact greedy allocation against the $600,000 notional limit, an enforcement pass that rebuilds the book to the exchange's specification, and split-half testing every information coefficient against a 200-shuffle permutation null at a Bonferroni threshold.

Susquehanna International Group (SIG) Algothon, Algorithmic Trading Competition 2026
Team Lead | Top 15% of all entrants, Australia and New Zealand | Python, NumPy, JavaScript, HTML/CSS

- Led development of a market neutral trading engine that achieved $262,508 cumulative PnL on 250 held out trading days, with 29ms median decision latency across 51 instruments
- Combined several quantitative strategies, vector auto regression, cointegration pairs, and PCA based cross sectional reversals with daily portfolio weight re estimation
- Built a lightweight browser based monitoring terminal using Vanilla.js, HTML Canvas, CSS so we could monitor individual trading days and inspect alpha signals, drawdown, Sharpe and hit rate
- Added automated reconciliation checks to ensure the dashboard matched the competition grader accurately along with headless Node.js smoke tests covering the main charting paths
- tested more than 100 candidate signals, including LSTMs, Kalman filters, and multi-asset cointegration models, and narrowed them down using separate training and holdout datasets

PROJECTS

Efficient Email Filtering System, Bloom Filter | **C++**

- Built a Bloom Filter from scratch to detect duplicate emails with 0 false negatives and a 0.8% false positive rate on a 1,485 item workload.
- Implemented seven hash functions over 15,000 bit array and added Base64 serialization
- Turned the project into a usable command line tool with a file and live ingestion, supporting real time ADD, CHECK and SHOW operations

Intraday Volatility Forecasting, Optiver Industry Project | **R, rugarch, glmnet, data.table**

- Cut QLIKE forecast error 38% against the HAR-X baseline, by specifying a GJR-GARCH model that won 33 of 40 stock horizon cells across 10 US equities.
- Produced 48,600 out of sample forecasts with zero failed fits across 360 backtest simulations and cut QLIKE from 1.38 to 0.181 between worst and best case, by caching period level features over nine months of Optiver limit order book data and rolling and expanding windows from 30 to 500 periods.